

VIRAN LINGALA

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EDUCATION

Clemson University, Clemson, SC

Master of Science in Computer Engineering

Aug 2023 – May 2025

CGPA: 3.86 / 4.0

Kakatiya Institute of Technology and Science, Warangal, India

Bachelor of Technology in Electronics Communication and Instrumentation Engineering

Aug 2019 – May 2023

CGPA: 3.4 / 4.0

EXPERIENCE

Software Automation Engineer (Quantum)

LPS Qubit Collaboratory (LQC), University of Maryland

May 2025 – Present

- Designed and developed **automation software** for quantum measurement systems, building Python-based instrument drivers to control **VNAs, Spectrum Analyzers, RF Sources, and Lock-in Amplifiers** for **Josephson Junction, PPMS Transport, and Resonator** measurements.
- Automated experimental workflows with real-time **data acquisition, synchronization, and remote control**, improving reproducibility and multi-user accessibility in cryogenic quantum setups.
- Developed **AI integrated lab management tools** for project and resource management using **Flask, JavaScript, SQL, and HTML/CSS**, integrating task ticketing, equipment scheduling, and team coordination dashboards.
- Implemented **FPGA-based control and measurement systems** using the **QubiC platform**, enabling low-latency feedback and scalable automation for quantum device characterization.

Software Automation Intern (Quantum)

LPS Qubit Collaboratory (LQC), University of Maryland

May 2024 – Aug 2024

- Assisted in developing **driver-level control software** for quantum hardware using Python, ensuring reliable TCP/IP communication, instrument control, and seamless integration with the automation frameworks.
- Designed and deployed a secure **local network infrastructure** interconnecting instruments and controllers, and enhanced internal workflows by integrating features from **Quantify-Core**, improving system reliability and data acquisition efficiency.

AI/ML Research Assistant

Department of Physics and Astronomy, Clemson University

Nov 2023 – May 2024

- Designed and developed an **AI agent** using **RAG** with **OpenAI APIs** and **Hugging Face embeddings**, integrated with a **FAISS vector database** for efficient document retrieval and semantic understanding.
- Automated extraction of **nanofabrication process data** from scientific literature by integrating **OCR-based text scraping** and **image detection** for scientific diagrams, transforming unstructured papers into structured, relational datasets categorized by material and fabrication techniques.
- Implemented and **fine-tuned Transformer-based embeddings** on domain-specific (Nano-Fabrication) text to enhance contextual understanding and improve relational data extraction accuracy.
- Deployed the end-to-end application using a **CI/CD workflow** with **GitHub, Heroku, and Azure DevOps**, connected to a **cloud-hosted MySQL database** for real-time data retrieval through an interactive web interface.

ML Engineer *Accenture, India*

Nov 2022 – Aug 2023

- Designed and deployed an **Employee Attrition Prediction System** using **Python, Scikit-learn, XGBoost, and Flask**, integrating data cleaning, feature engineering, and model optimization pipelines to achieve high predictive accuracy for real-time workforce turnover forecasting.
- Integrated the trained model into a **Flask-based dashboard** and deployed it on **Azure Cloud**, automating data ingestion, inference, and visualization for actionable business insights.

TECHNICAL SKILLS

Programming

Python, C++, Verilog, VHDL, MATLAB, JavaScript, HTML/CSS, Flask

AI/ML

Pandas, NumPy, Matplotlib, Data Visualization, Scikit-learn, Hugging Face Transformers, OpenAI Embeddings, LangChain, FAISS, Deep Learning, OCR, Feature Engineering, Data Analysis

Quantum Computing & FPGA Systems

FPGA Development, QubiC FPGA Platforms, RTL Design, Embedded Systems, LabVIEW, Jetson Nano, Josephson Junctions, PPMS, Resonators, Quantify-Core, Real-Time Data Acquisition

Cloud & DevOps Tools

MySQL, Azure Cloud, Linux, Docker, Git, Azure DevOps, TCP/IP Networking, Socket Programming, Local Network Configuration, CI/CD Pipelines, Cloud Deployment

Control Instruments

Vector Network Analyzers (VNA), Spectrum Analyzers, Lock-in Amplifiers, RF Signal Sources, Oscilloscopes, Arbitrary Waveform Generators (AWG)